Introduction to Pharmacoepidemiology

Study of the use and effects of drugs in large populations.



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Definition

Study of the use and effects of drugs in large populations.

Purpose

Bridge between clinical pharmacology and public health.

Why it matters

Helps identify real-world effectiveness, safety, and cost-benefit of medications.

Applications

Post-marketing surveillance, drug policy, rational prescribing.

Incidence vs Prevalence

Incidence

New cases in a defined period ÷ population at risk.

Prevalence

All existing cases ÷ total population at a specific time.

Example:

- 50 new asthma cases in 1 year among 1 000 people → **Incidence = 0.05**.
- 200 total asthma cases → **Prevalence = 0.20**.

Key idea: Incidence shows risk, prevalence shows burden.

Relationship Between Incidence Rate and Prevalence

$$P = I \times D$$

(Concept only – no exam numbers).

When disease duration ↑, prevalence increases even if incidence is constant.

Chronic diseases (e.g., diabetes): long duration → high prevalence.

Acute diseases (e.g., flu): short duration → low prevalence.



Competing Risks & Cumulative Incidence

Competing risk

Event that prevents the outcome of interest from occurring.

Example: death from heart disease before developing cancer.

Cumulative incidence

Probability of developing disease over time, adjusted for competing risks.

Ignoring competing risks **overestimates** disease probability.

Relative Risk (RR)

1

Definition

Risk in exposed ÷ risk in unexposed.

2

Formula

RR = (a / (a + b)) ÷ (c / (c + d)).

3

Example

30 / 200 exposed vs 10 / 300 unexposed → **RR ≈ 4.5**.

4

Interpretation

Exposure group has **4.5× higher risk**.

RR > 1: risk ↑

RR = 1: no association

RR < 1: protective.

Attributable Risk (AR) & Population AR

AR (exposed)

Risk exposed − Risk unexposed.

Population AR

Total risk − risk in unexposed fraction.

Example:

RiskE = 0.15 RiskU = 0.05 → **AR = 0.10 (10 %)**.

Meaning: 10 % of disease among exposed is due to exposure.

Use: Quantifies potential benefit if exposure eliminated.

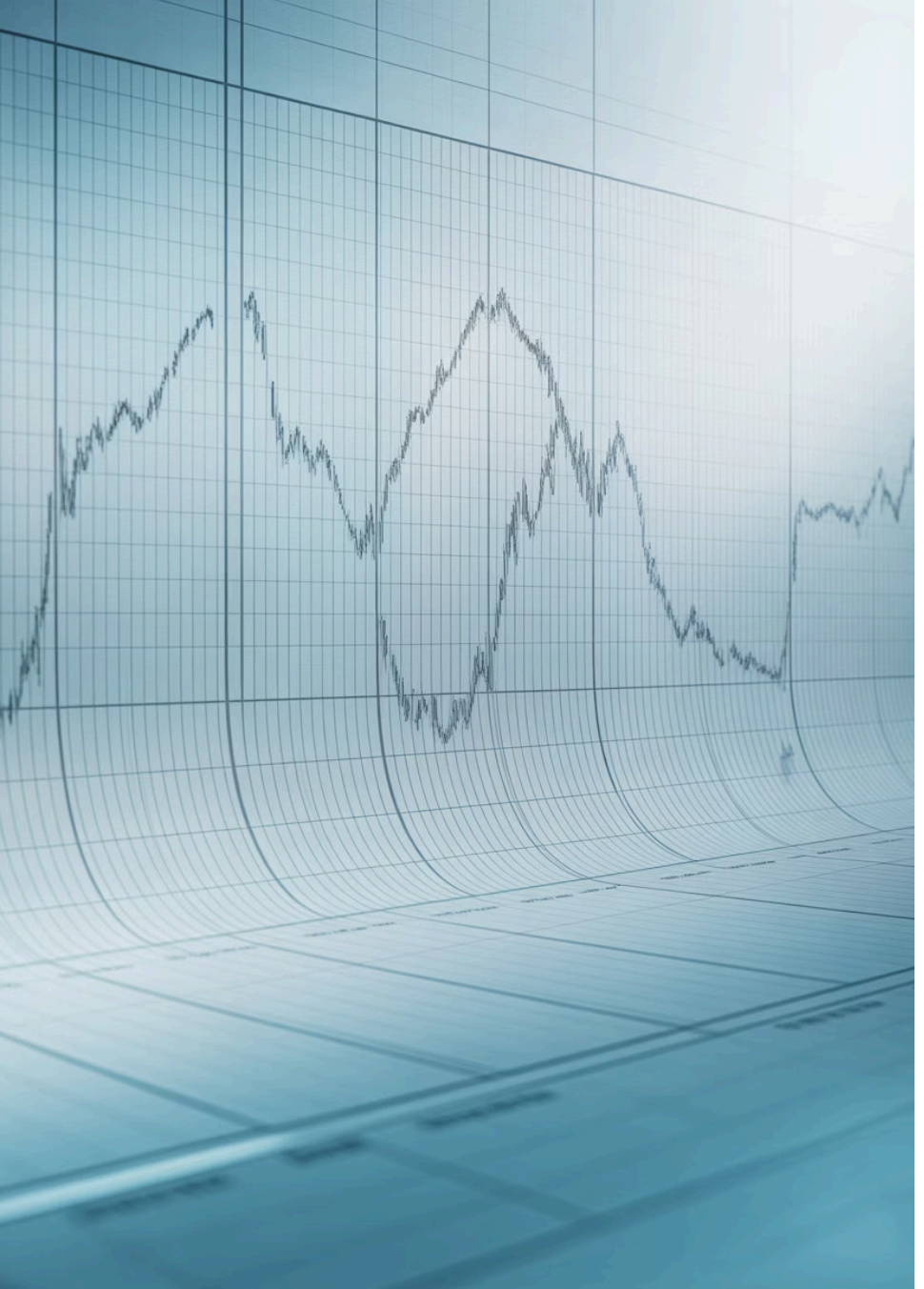
Odds Ratio (OR) vs Relative Risk

Feature	RR	OR
Study Type	Cohort / Trial	Case-Control
Measures	Probability	Odds
Use When	Incidence known	Rare diseases
Interpretation	1 = no effect	1 = no effect

Example (new):

Odds of smoking in cases = 0.6; in controls = 0.2 → **OR = 3.0**.

Interpretation: Smokers have 3× the odds of disease.



Hazard Ratio (HR) and Time-to-Event

Definition

Ratio of hazard (instantaneous risk) between groups.

Example

HR = 0.70 means **30 % reduction** in event rate.

Used in

Survival analysis (Kaplan–Meier curves, Cox model).

Advantage

Considers both event occurrence and time until event.

Number Needed to Treat (NNT) & Number Needed to Harm (NNH)

NNT

NNT = 1 / ARR (Absolute Risk Reduction).

NNH

NNH = 1 / AR (Absolute Risk Increase).

Example:

Event rate 10 % (control) vs 6 % (treated) → ARR = 0.04 → **NNT = 25**.

Interpretation: Treat 25 patients → prevent 1 event.

Smaller NNT = more effective; larger NNH = safer.

Confidence Intervals (CI) & Significance

95 % CI: range likely containing true value.

If CI for RR/OR includes 1 → not significant.

Example:

OR = 1.8 (1.1 – 2.3)

→ significant.

OR = 1.8 (0.9 – 2.8)

→ not significant.

CI shows both precision and certainty.

Bias in Research



Selection bias

Participants differ from population (volunteer bias).



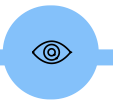
Information bias

Measurement errors (exposure misclassified).



Recall bias

Cases remember past exposure differently than controls.



Observer bias

Researcher expectation influences assessment.

Confounding and Control Methods

Confounding

Distortion caused by a third variable linked to both exposure and outcome.

Example: Age confounds smoking and lung disease.

Control strategies:

- Randomization
- Matching (age, gender)
- Stratification
- Multivariate analysis (adjusted OR/RR).



Loss to Follow-Up & Study Validity



Definition

Participants drop out before study ends.



Consequences

Reduced sample size, biased results, lower power.



Prevention

Good communication, reminders, participant tracking, incentives.



Reporting

Always state % lost to follow-up.

Cohort Studies

Prospective

Follow from exposure → outcome.

Retrospective

Use records of past exposures.

Advantages:

- Measure incidence & RR
- Clear temporal relation

Limitations:

- Time consuming
- Costly
- Loss to follow-up

Case-Control & Nested Case-Control

Case-Control

Start with disease status → look back for exposure.

Nested design

Cases and controls drawn from an existing cohort.

Advantages:

- Efficient for rare diseases
- Lower cost
- Quick

Key concept: Matching controls reduces confounding.

Cross-Sectional & Ecological Studies

Cross-Sectional

Snapshot of exposure and disease at one time.

Ecological

Uses group data (population-level).

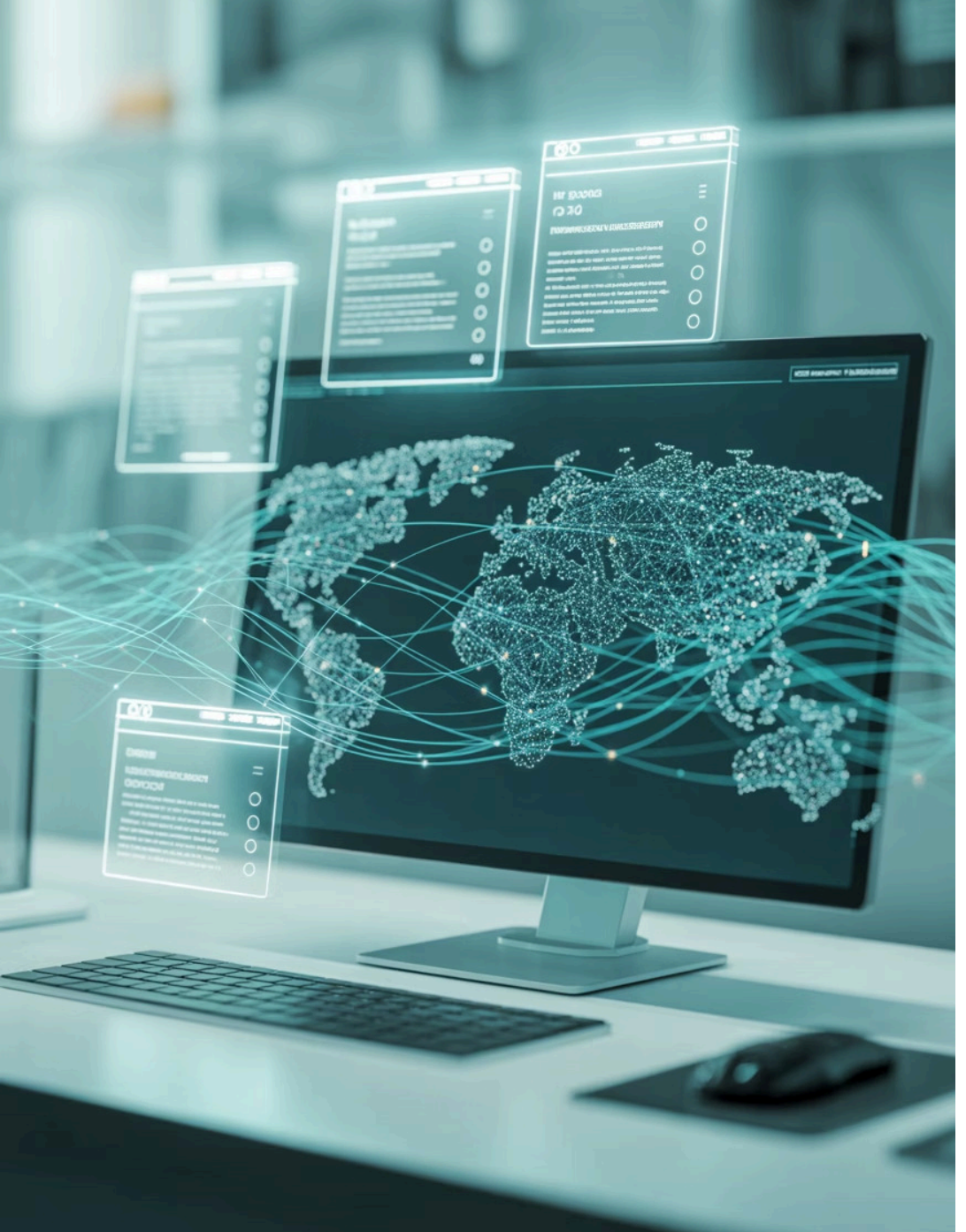
Limitation: Ecological fallacy – group association ≠ individual association.

Use: Generating hypotheses, not causal proof.

Adverse Drug Reactions (ADR) Types

Type	Nature	Example	Predictable?
A	Dose-dependent, common	Hypotension from antihypertensive	Yes
B	Idiosyncratic / immune	Anaphylaxis to penicillin	No
C	Chronic use effects	Corticosteroid osteoporosis	Partly
D	Delayed effects	Carcinogenesis after years	No

Monitoring reduces morbidity and cost.



Pharmacovigilance & Post-Marketing Surveillance

Definition

Science and activities relating to detection, assessment, understanding and prevention of adverse effects.

Phase IV studies

Detect rare / long-term effects in real populations.

Systems

SFDA National Center for Pharmacovigilance, WHO-UMC VigiBase.

Goal

Continuous drug safety and benefit-risk assessment.

Developing a Strong Research Question

FINER criteria

- Feasible
- Interesting
- Novel
- Ethical
- Relevant

PICO format

- Population
- Intervention
- Comparison
- Outcome

Hypothesis: A testable prediction derived from the question.

Good example: "Does AI-based monitoring reduce medication errors in ICU patients?"